

Data Science

CSE558

Question

1. In a d -dimensional space, what is the size of the largest set of mutually orthogonal unit vectors?
2. In a d -dimensional space, what is the size of the largest set of mutually nearly orthogonal unit vectors, i.e., $|\langle x_i, x_j \rangle| < 0.01$ with 0.99 probability?

$O(d)$

$O(\log d)$

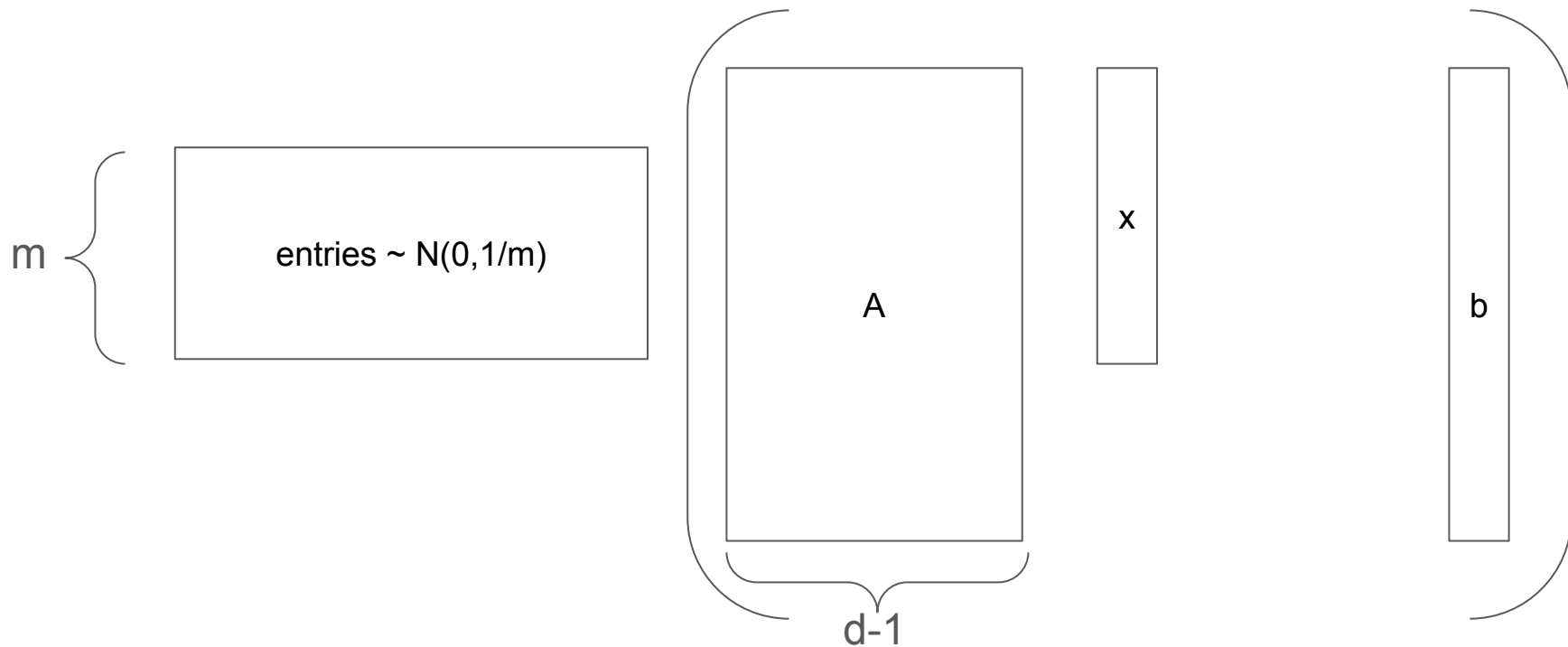
$O(d^2)$

$O(2^d)$

Overview

- Fast Linear Regression
- Low Rank Approximation

Fast Linear Regression with JL

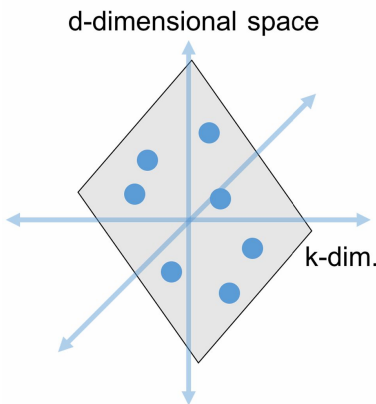


Overview

- Fast Linear Regression
- Low Rank Approximation

Low Rank Approximation

Suppose $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ lie on a k -dimensional space.



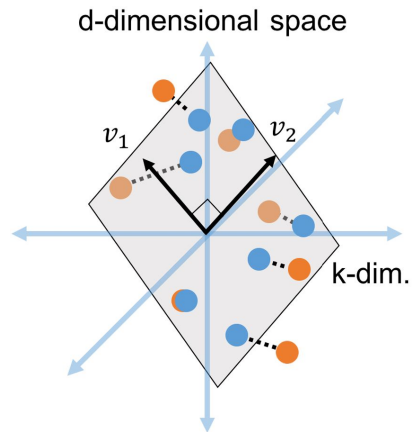
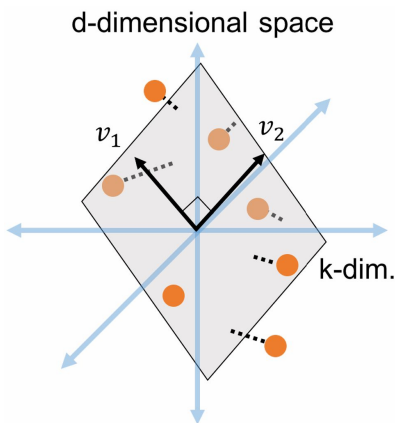
For all x_i there exists no distortion embedding $c_i \in \mathbb{R}^k$ such that $\|x_i - x_j\| = \|c_i - c_j\|$.

Proof:

Is this a unique embedding?

Low Rank Approximation

Suppose $x_1, x_2, \dots, x_n \in \mathbb{R}^d$ close to a k -dimensional space.



For every x_i a distorted embedding $c_i \in \mathbb{R}^k$ is a good approximation.

- Compute embedding
- For every x_i compute approximate $y_i \in \mathbb{R}^d$ (Low Rank Approximation)